

Sean Luka Girgis

Senior Data Engineer | Python, SQL, AWS Data Pipelines & Production Data Operations

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Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, AWS data platforms, ETL/ELT pipelines, cloud data operations, data ingestion, transformation, validation, data quality, lineage-style documentation, monitoring, CI/CD concepts, and production support. Strong background designing and maintaining enterprise-scale data workflows, reconciling pipeline outputs, optimizing SQL/reporting layers, documenting data flows, and supporting reliable operational analytics in regulated enterprise environments. Best aligned to remote data engineering roles requiring Python, SQL, cloud or hybrid data operations, orchestration, validation, monitoring, data reliability, and production pipeline support; GCP/Dataflow, formal public-sector delivery, and cloud certifications are positioned carefully as adjacent or non-claimed areas.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable ingestion, transformation, querying, and downstream analytics workloads.
- Supported production data operations in cloud and hybrid environments, including ingestion monitoring, transformation validation, reconciliation, reporting-layer support, and issue diagnosis.
- Developed Python and SQL workflows for data ingestion, cleansing, enrichment, standardization, validation, and delivery to analytics and business-facing data consumers.
- Built data quality, reconciliation, and validation practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve completeness, consistency, and trust in pipeline outputs.
- Documented data flows, transformation rules, metric definitions, validation checkpoints, lineage-style notes, operational runbooks, and support procedures to improve maintainability and auditability.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Collaborated with infrastructure, application, analytics, and business teams to troubleshoot production data issues, improve monitoring practices, and deliver reliable data products.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.

- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and data/reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, dashboards, operational analytics, and production reliability.
- Built Perl and KornShell/ksh automation for extracting, transforming, validating, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed reusable scripts, dashboards, alert logic, SLA thresholds, technical documentation, and operational standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze telemetry, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and data feed processes for telecom-scale transaction systems.

- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and operational housekeeping.
- Implemented business logic in billing and integration workflows using C/C++, Pro*C, PL/SQL, SQL scripts, XML, shell automation, and enterprise messaging interfaces.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine

Built Python/Pandas/PySpark workflows to process infrastructure telemetry, validate pipeline outputs, prepare forecasting datasets, and deliver capacity insights through repeatable reporting and dashboard workflows.

Technologies: Python, Pandas, PySpark, SQL, Prophet, scikit-learn, Data Validation

AI-Powered Job Search Pipeline

Built a Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking.

Emphasized structured data processing, validation, workflow reliability, and documentation.

Technologies: Python, JSON, FAISS, Sentence Transformers, Grok / xAI API, Quality Gates